

Exam FM Formula Sheet

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Every formula below is derived and explained in full in the Exam FM Study Manual. This page exists purely for fast lookup and final-week memorization, condensed to fit a short printout. Nothing here is a substitute for understanding where each formula comes from.

1. Time Value of Money (5–15%)

COMPOUND ACCUMULATION

$$a(t) = (1+i)^t, AV = PV(1+i)^n$$

DISCOUNT FACTOR

$$v = 1/(1+i), PV = AV \cdot v^n$$

SIMPLE INTEREST

$$a(t) = 1 + it$$

EFFECTIVE DISCOUNT RATE

$$d = iv = 1 - v = i/(1+i)$$

NOMINAL RATE CONVERSION

$$1+i = (1+i^{(m)}/m)^m = (1-d^{(m)}/m)^{-m} = e^{\delta}$$

FORCE OF INTEREST (CONSTANT)

$$\delta = \ln(1+i)$$

FORCE OF INTEREST (VARIABLE)

$$\delta(t) = a'(t)/a(t), a(t) = \exp[\int_0^t \delta(s)ds]$$

EQUATION OF VALUE

$$\Sigma(\text{inflows disc./accum. to } T) = \Sigma(\text{outflows disc./accum. to } T)$$

Rate Conversion Quick-Reference

Given	To find i
$i^{(m)}$	$i = (1+i^{(m)}/m)^m - 1$
d	$i = d/(1-d)$
$d^{(m)}$	$i = (1-d^{(m)}/m)^{-m} - 1$
δ	$i = e^{\delta} - 1$

2. Annuities (20–30%)

ANNUITY-IMMEDIATE (PV, AV)

$$a_{n-} = (1-v^n)/i, s_{n-} = [(1+i)^n - 1]/i$$

ANNUITY-DUE (PV, AV)

$$\ddot{a}_{n-} = a_{n-}(1+i) = (1-v^n)/d, \ddot{s}_{n-} = s_{n-}(1+i)$$

LEVEL PERPETUITIES

$$a_{\infty} = 1/i \text{ (immediate)}, \ddot{a}_{\infty} = 1/d \text{ (due)}$$

ANNUITY PAYABLE M-THLY

$$a_{n-}^{(m)} = (1-v^n)/i^{(m)}, \ddot{a}_{n-}^{(m)} = (1-v^n)/d^{(m)}$$

CONTINUOUS ANNUITY

$$\bar{a}_{n-} = (1-v^n)/\delta$$

DEFERRED ANNUITY (M-YEAR DEFERRAL, N-YEAR IMMEDIATE)

$$PV = v^m \cdot a_{n-}$$

INCREASING ANNUITY-IMMEDIATE

$$(Ia)_{n-} = [\ddot{a}_{n-} - n \cdot v^n] / i$$

DECREASING ANNUITY-IMMEDIATE

$$(Da)_{n-} = [n - a_{n-}] / i$$

INCREASING PERPETUITY-IMMEDIATE

$$(Ia)_{\infty} = 1/i + 1/i^2 = (1+i)/i^2$$

GEOMETRIC ANNUITY (PAYMENTS $1, (1+k), (1+k)^2, \dots$)

$$PV = [1 - ((1+k)/(1+i))^n] / (i-k), i \neq k$$

GEOMETRIC PERPETUITY

$$PV = 1/(i-k), \text{ requires } i > k$$

3. Loans (15-25%)

LEVEL LOAN PAYMENT

$$\text{Principal} = \text{Payment} \times a_{n-}$$

OUTSTANDING BALANCE — PROSPECTIVE

$$B_t = \text{Payment} \times a_{(n-t)-}$$

OUTSTANDING BALANCE — RETROSPECTIVE

$$B_t = \text{Principal} \times (1+i)^t - \text{Payment} \times s_{t-}$$

INTEREST & PRINCIPAL SPLIT

$$I_t = i \times B_{(t-1)}, P_t = \text{Payment} - I_t$$

PRINCIPAL REPAID GROWTH

$$P_t = P_1 \times (1+i)^{(t-1)} \text{ (level-payment loan)}$$

4. Bonds (15-25%)

BOND PRICE

$$P = Fr \times a_{n-} + C \times v^n$$

PREMIUM / DISCOUNT

$$P - C = (Fr - Ci) \times a_{n-}$$

BOOK VALUE RECURSION

$$B_t = B_{(t-1)}(1+i) - Fr$$

PREMIUM AMORTIZED IN PERIOD T

$$Fr - i \cdot B_{(t-1)}$$

DISCOUNT ACCUMULATED IN PERIOD T

$$i \cdot B_{(t-1)} - Fr$$

Premium vs. Discount at a Glance

Relationship	Bond trades at	Book value over time
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$r > i$	Premium ($P > C$)	Amortizes down to C
$r < i$	Discount ($P < C$)	Accumulates up to C
$r = i$	Par ($P = C$)	Constant at C

5. General Cash Flows, Portfolios & ALM (20–30%)

DOLLAR-WEIGHTED RETURN (EXACT)

$$B(1+i) - \sum C_t(1+i)^{(1-t)} = A$$

DOLLAR-WEIGHTED RETURN (APPROX.)

$$i \approx I / [B + \sum C_t(1-t)]$$

TIME-WEIGHTED RETURN

$$1+i = \Pi(\text{Value just before each CF} / \text{Value just after previous CF})$$

FORWARD RATE FROM SPOT RATES

$$(1+s_t)^t(1+f_{[t,t+k]})^k = (1+s_{(t+k)})^{(t+k)}$$

MACAULAY DURATION

$$D_{\text{Mac}} = \sum t \cdot \text{PV}(\text{CF}_t) / \sum \text{PV}(\text{CF}_t)$$

MODIFIED DURATION

$$D_{\text{mod}} = D_{\text{Mac}} / (1+i)$$

CONVEXITY

$$C = \sum t(t+1) \cdot \text{PV}(\text{CF}_t) / [\sum \text{PV}(\text{CF}_t) \times (1+i)^2]$$

1ST-ORDER APPROX. (MODIFIED DURATION)

$$\Delta \text{PV} / \text{PV} \approx -D_{\text{mod}} \times \Delta i$$

1ST-ORDER APPROX. (MACAULAY DURATION)

$$\Delta \text{PV} / \text{PV} \approx -D_{\text{Mac}} \times \Delta i / (1+i)$$

2ND-ORDER (CONVEXITY-ADJUSTED)

$$\Delta \text{PV} / \text{PV} \approx -D_{\text{mod}} \times \Delta i + \frac{1}{2} \times C \times (\Delta i)^2$$

Redington Immunization — Three Conditions

#	Condition
1	$\text{PV}(\text{Assets}) = \text{PV}(\text{Liabilities})$
2	$\text{Duration}(\text{Assets}) = \text{Duration}(\text{Liabilities})$
3	$\text{Convexity}(\text{Assets}) > \text{Convexity}(\text{Liabilities})$

Immunization / Matching Techniques Compared

Technique	Protects against	Requires
Redington immunization	Small parallel shifts in i	PV, duration, convexity conditions above
Full immunization	Any size shift in i	Asset CF strictly before & after each liability CF ("sandwich")
Cash-flow matching / dedication	Any shift in i (no reinvestment risk)	Exact asset CF for every liability CF

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Exam FM Formula Sheet — condensed from the Exam FM Study Manual's Appendix A. For derivations and worked examples, see the full study manual.